

Attention Lecture 4

2024-09-09

Lecture 3 student question

I'm a bit confused after that lecture. If i walk into a room and just look at nothing in particular, am I bottlenecking a particular feature, or am i simulatenously taking everything in?

You are:

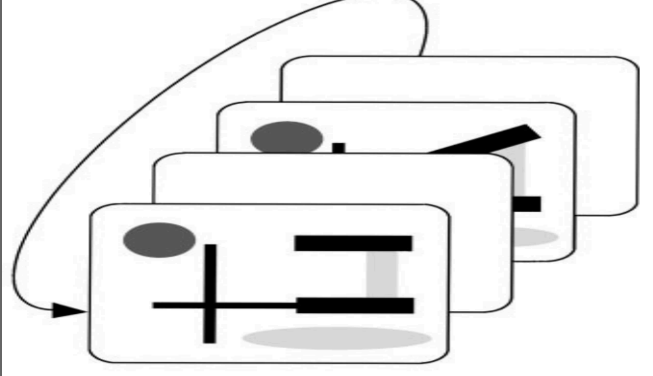
- Simultaneously processing color, motion, but not their combination
- Extensively processing (through the bottleneck) one or a few objects at the location of your attention
- Doing additional processing that we don't understand through your non-selective pathway

Combining features within a dimension

[globalShape.html](#)

CHANGE BLINDNESS - Chapter 6

Points to “a grand illusion of visual experience”?



Blank-screen sandwiches: Lessons

- Grand illusion of visual experience?
- Peripheral vision
- Motion detectors
 - Blank screen
 - Gradual changes
- Memory

Find the change





Change blindness

Be able to explain why it inspired the “grand illusion of visual experience” theory



It feels like we see everything at once

- But, change blindness!
- So, we're wrong that we see everything simultaneously.
- There's a "grand illusion of visual experience"
- It's like the refrigerator light illusion

Argument for “the grand illusion”

- **Premise 1.** If you visually experience most of a scene’s objects, you will notice major changes to them.
- **Premise 2.** You don’t detect major changes to most objects in a scene (in the blank-screen sandwich movies).
- **Conclusion.** Therefore, you *don’t* visually experience most objects in a scene.

If A, then B.

B is not true.

Therefore, A is not true.

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Answer a question about the “grand illusion of visual experience” argument.

Argument for “the grand illusion”

Premise 1. If you visually experience most of a scene’s objects, you will notice major changes to them.

Premise 2. You don’t detect major changes to most objects in a scene (in the blank-screen sandwich movies).

Conclusion. Therefore, you don’t visually experience most objects in a scene.

Needed to detect a change

1. Internal representations of the object before and after the change.
 2. A process that compares the before representation to the after.
 3. A process that calls attention to, or brings into conscious awareness, the comparison results.
- **Premise 1.** *If you visually experience most of a scene's objects, you will notice major changes to them.*
 - Does having *visual experience of all the objects in a scene* mean you also have 1 and 2?

The processes for detecting a change

1. Before / after

2. Comparison

Premise 1. If you visually experience most of a scene's objects, you will notice major changes to them.

- Premise 1 is questionable
- You might be able to have visual experience of the objects without having 1 and/or 2

Argument for the grand illusion

Premise 1. If you visually experience most of a scene's objects, you will notice major changes to them.

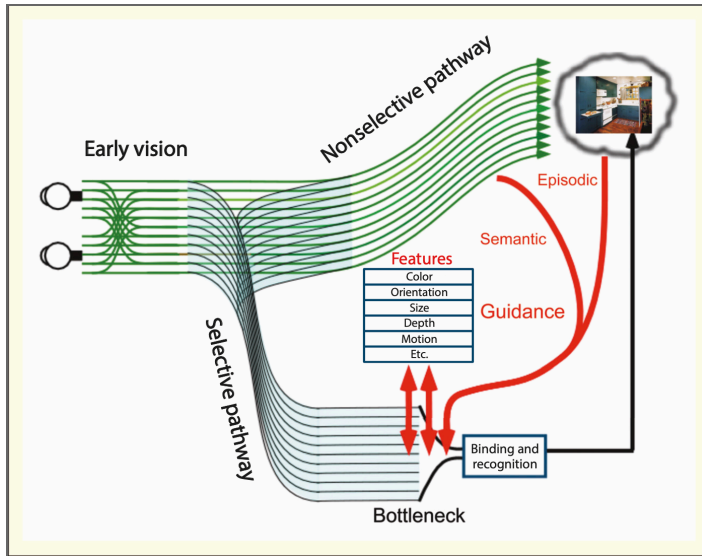
Premise 2. You don't detect major changes to most objects in a scene (in the blank-screen sandwich movies).

Conclusion. Therefore, you don't visually experience most objects in a scene.

- Premise 1 was part of the argument for the grand illusion
- Premise 1 is highly questionable
- The argument for the “grand illusion” is not good!
- Maybe we do experience the whole scene

Re-run Menti grand illusion argument poll

Maybe we do experience the whole scene



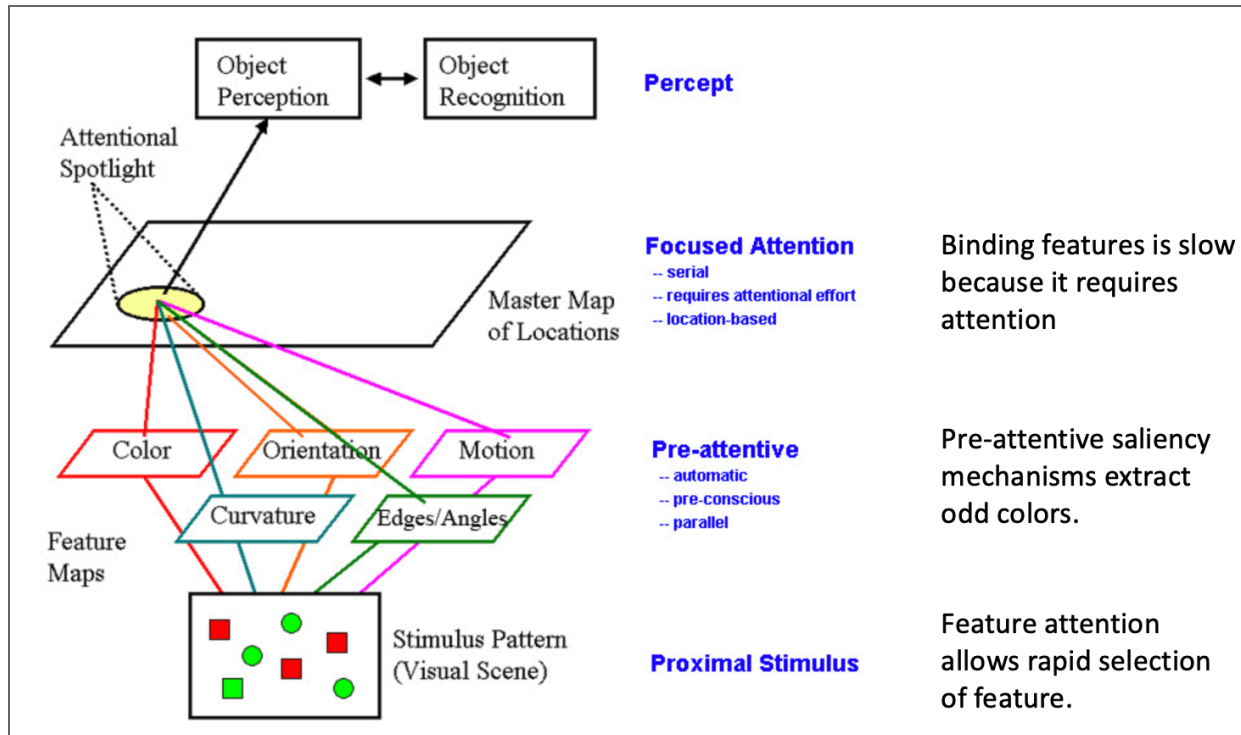
Review of the argument (now in essay format in case that helps)

The “grand illusion” argument: if we really did experience everything in the scene, we would quickly detect changes to the objects in the scene. So, because we don’t quickly detect many changes to the scene (change blindness), we must *not* really experience everything.

But the grand illusion argument is actually *not* a great argument; the first part that says that if we really did experience everything, we would quickly detect changes. Maybe we really do experience everything but we don’t have a comparison process that’s always comparing everything. Or maybe we really do experience everything but we don’t have a representation of the scene a short time ago, so we can’t do the comparison. In other words, the idea that “if we really did experience everything, we would quickly detect changes” may be false.

Explaining blank screen sandwiches

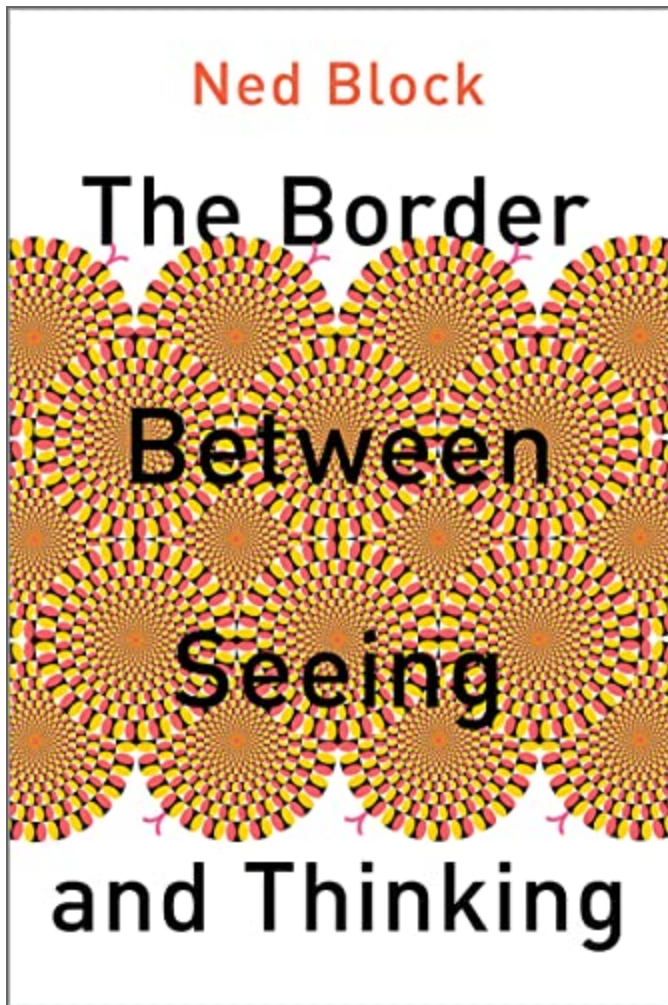
- Focused attention is needed to detect change



Change blindness -> “the grand illusion”

- You can still believe there is a “grand illusion” of visual experience, but change blindness is not a good argument for it.

For more, see Overgaard (2018) or read Block’s 550-page book.



Blank-screen sandwiches

- Peripheral vision
- Motion detectors
- Memory

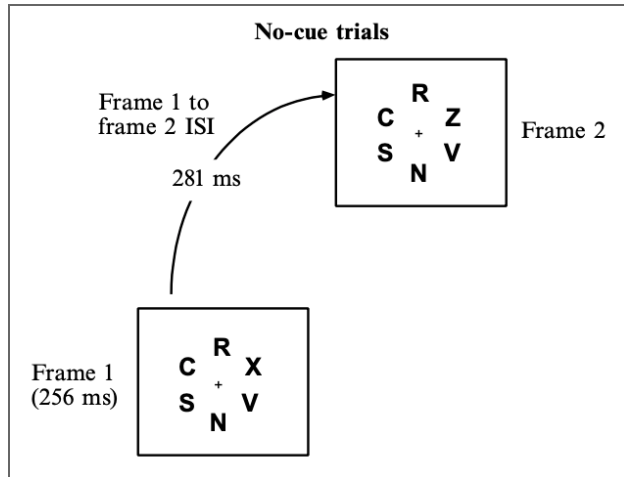


Peripheral vision

Good experimenters ensure that the object is visible in the periphery.

- If the object is not visible in the periphery, need *overt* attention to the location, not just *covert* attention
- Unsurprising that this can't be done at all locations at once!

Poor peripheral vision



- In rigorous experiments, stimuli are placed carefully so they are visible in the periphery

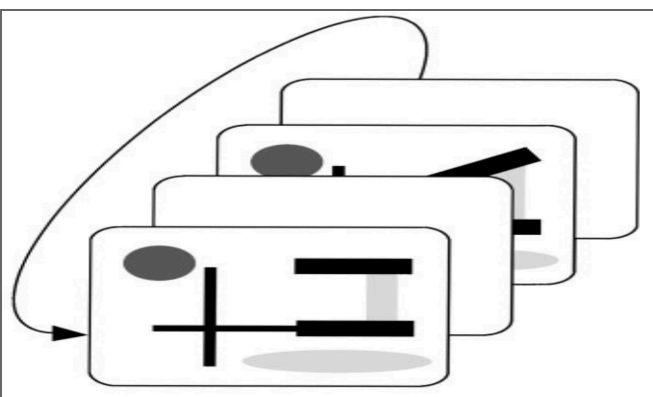
Becker, Anstis, and Pashler (2000).

Once you find it, it's hard to un-see it



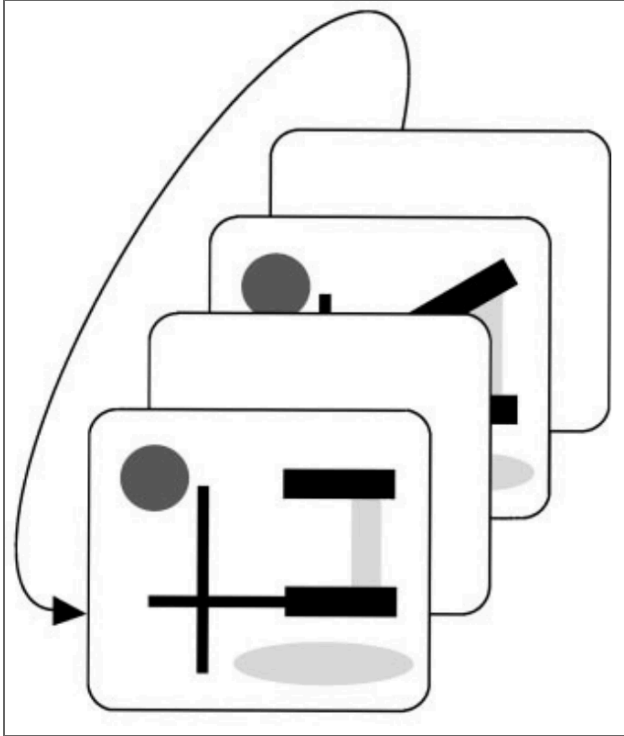
- Memories guide attention quickly
- This helps us in countless situations, including on the road

Blank screen sandwiches



No blank frame

Find the change, but this time there will be no blank screen



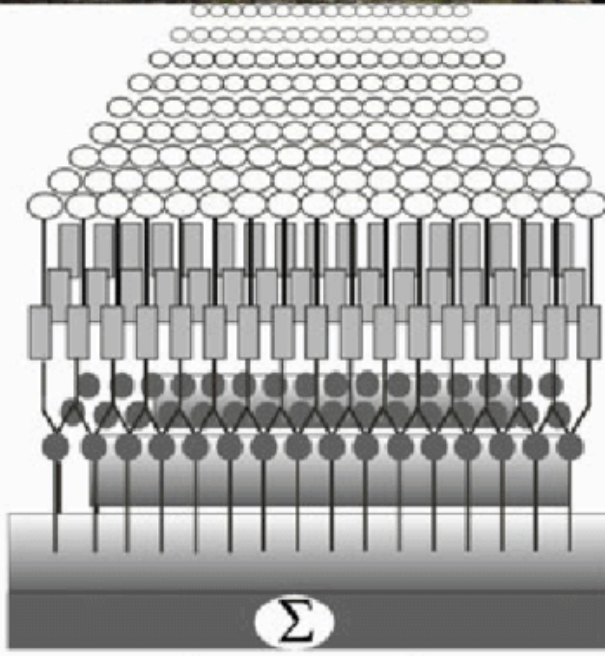
No blank frame



a

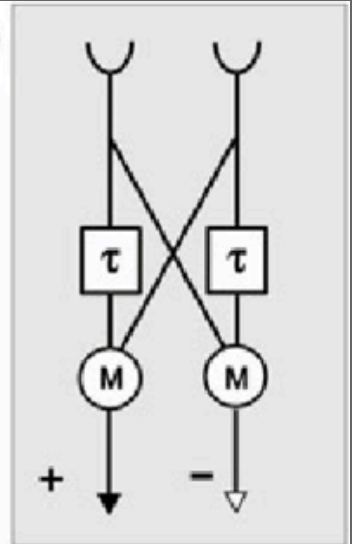


photoreception
spatio-temporal
filtering
motion detection
spatial pooling
wide-field neurons



control of
behavioral components

b





- Motion detectors respond to flicker
- If there's motion everywhere, there might as well be motion nowhere
- Odd motion detector response no longer makes change location salient

See responses to last Menti



- “Mud splashes” almost as good as blank screen
- Motion detectors misdirect attention

What made the mud splashes more salient than the white road lines?

- They are bigger and thus stimulate the motion detectors more strongly
- They are more different to their surround (parallel bottom-up attention processing)

Gradual changes

Video player configuration error

Error 153

Watch video on YouTube

Youtube

Gradual changes

Why are gradual changes hard to find?

- They don't stimulate motion detectors

Blank-screen sandwiches: Lessons

- Grand illusion of visual experience?
- Peripheral vision
- Motion detectors
 - Blank screen
 - Gradual changes
- Memory

Tutorial Week 5

Implications for driving

Change blindness in continuous movie

Video player configuration error

Error 153

Watch video on YouTube

Youtube

Video player configuration error

Error 153

[Watch video on YouTube](#)

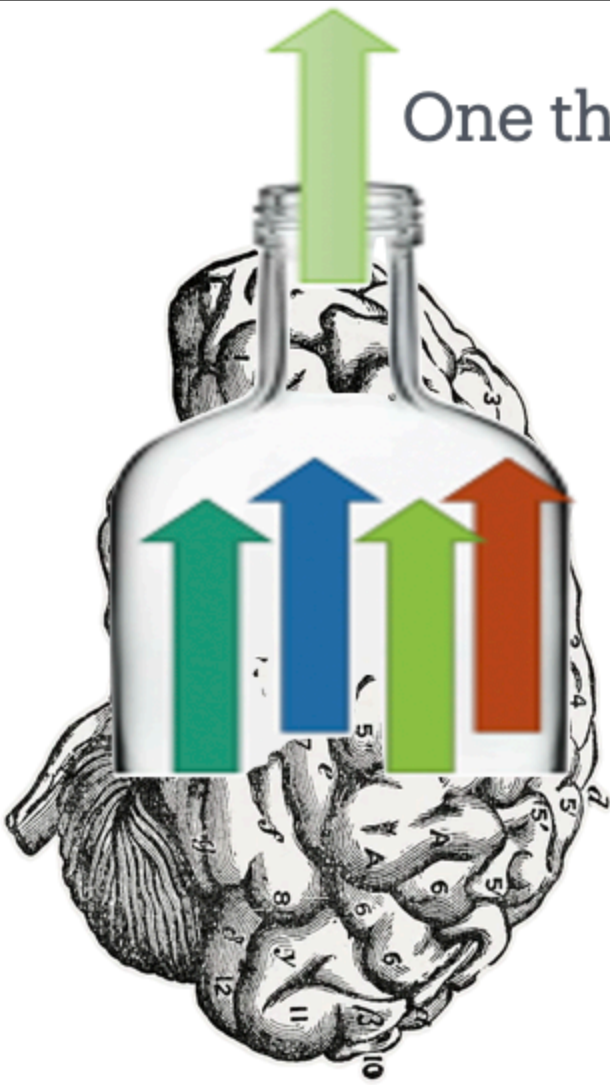
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Why do people miss many of the changes?

- Did you notice that this slide has no background image? 😊

One thing

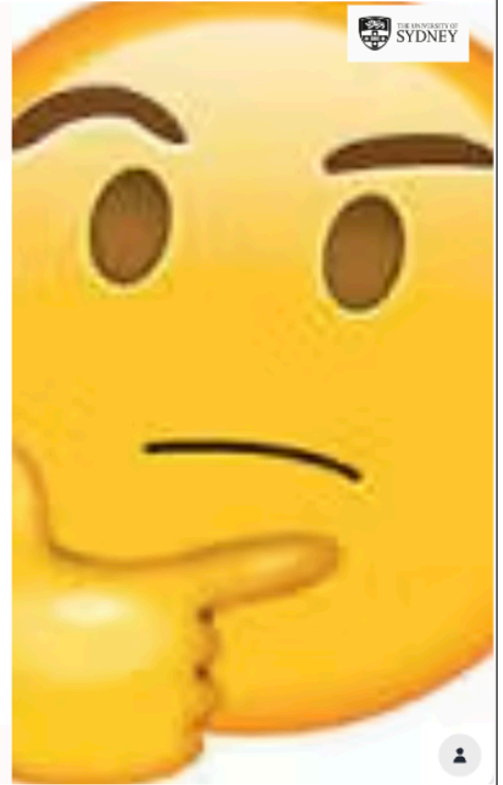
Many things



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Thing(s) I didn't
understand

Waiting for responses ...



PSYC2016: Attention

Speaker notes